

Paeonol Derivative, 6'-Methyl Paeonol, Attenuates A β -Induced Pathophysiology in Cortical Neurons and in an Alzheimer's Disease Mice Model

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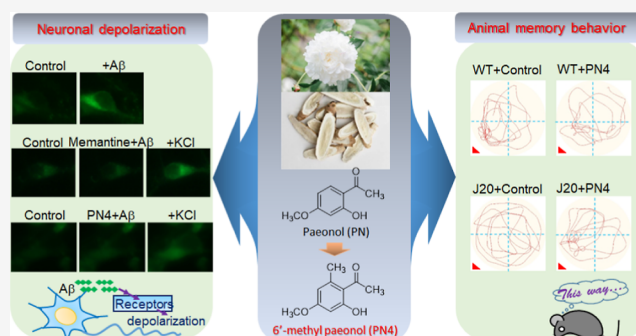
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ABSTRACT: Herbs themselves and various herbal medicines are great resources for discovering therapeutic drugs for various diseases, including Alzheimer's disease (AD), one of the common neurodegenerative diseases. Utilizing mouse primary cortical neurons and DiBAC4(3), a voltage-sensitive indicator, we have set up a drug screening system and identified an herbal extraction compound, paeonol, obtained from *Paeonia lactiflora*; this compound is able to ameliorate the abnormal depolarization induced by A β 42 oligomers. Our aim was to further find effective paeonol derivatives since paeonol has been previously studied. 6'-Methyl paeonol, one of the six paeonol derivatives surveyed, is able to inhibit the abnormal depolarization induced by A β oligomers.

Furthermore, 6'-methyl paeonol is able to alleviate the NMDA- and AMPA-induced depolarization. When a molecular mechanism was investigated, 6'-methyl paeonol was found to reverse the A β -induced increase in ERK phosphorylation. At the animal level, mice injected with 6'-methyl paeonol showed little change in their basic physical parameters compared to the control mice. 6'-Methyl paeonol was able to ameliorate the impairment of memory and learning behavior in J20 mice, an AD mouse model, as measured by the Morris water maze. Thus, paeonol derivatives could provide a structural foundation for developing and designing an effective compound with promising clinical benefits.

KEYWORDS: Alzheimer's disease, depolarization, ERK, Morris water maze, paeonol



INTRODUCTION

Alzheimer's disease (AD) is accompanied by two hallmarks: tau hyperphosphorylation and amyloid- β (A β) plaque formation.¹ In the aspect of A β pathophysiological state, A β oligomers, formed from monomers, are generally regarded as toxic molecules within A β structures such as A β fibrils and/or plaques.^{2,3} Currently, it is believed that A β increases the glutamate concentration within the synapse as well as outside in the cleft.^{4,5} This then leads to glutamate-induced neurotoxicity. A β could also mediate its activity via dysregulation of the NMDAR,^{6,7} the AMPAR,^{8,9} Kainate receptors,¹⁰ and/or mGluR receptors¹¹ resulting in a series of signaling defects and/or giving rise to tau hyperphosphorylation.^{12,13} Eventually, the functioning of signaling molecules, such as ERK, JNK, p38, GSK3 β , CaMKII, and CREB, is impaired, resulting in neuronal death.¹⁴

The A β hypothesis has been challenged from 2020 onward based on the failure of several clinical trial drugs that targeted the formation of A β into a higher order structure.^{15–17} Although this gap between the theory and clinical application over the past decade needs to be further investigated, neuronal

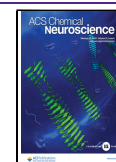
death caused by A β , as well as the effects of A β higher order structures, such as A β oligomers, seems to have already been verified and is reproducible, particularly the effect on glutamate receptors.^{18–21} Interestingly, aducanumab and lecanemab, two antibody drugs able to prevent A β monomers from forming a complex structure and/or bring about A β clearance, were approved in 2021/2023 by the FDA. Despite the ambiguity regarding the A β hypothesis, memantine has long been used to treat moderate to severe AD patients. This clinical therapeutic drug is an uncompetitive inhibitor of NMDAR.²² When the extra-synapse concentration of glutamate is raised by A β oligomers, the glutamates enter neurons via NMDAR2A, as well as via NMDAR2B.²³ Memantine is able to prevent calcium influx via NMDAR and thus block its downstream

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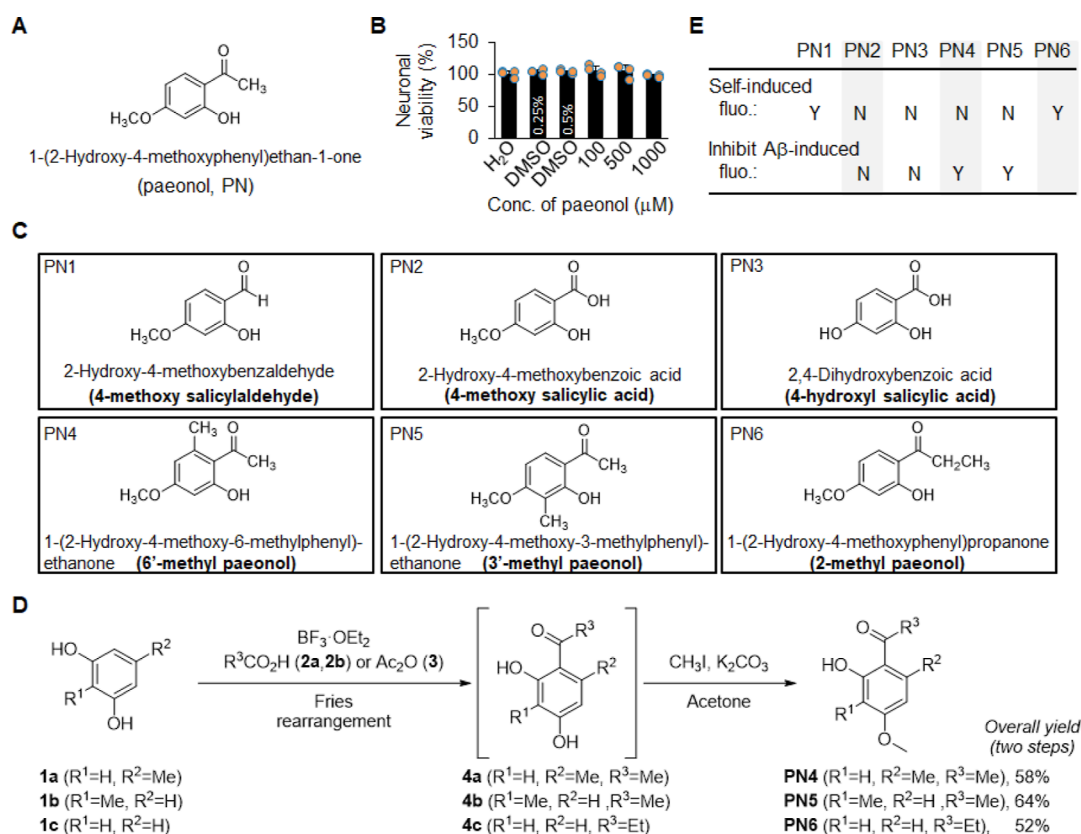


Figure 1. Structure of paeonol and relevant derivatives. (A) Structure of paeonol. (B) MTT assay using paeonol on primary cortical neurons. (C) Structure of various paeonol derivatives PN1~PN6. (D) Procedures used for synthesis of the paeonol derivatives PN4, PN5, and PN6. (E) Summary of the effects of paeonol and its derivatives on self-induced depolarization and on the inhibition of Aβ-induced depolarization. PN, paeonol; PN1, 4-methoxy salicylaldehyde; PN2, 4-methoxy salicylic acid; PN3, 4-hydroxyl salicylic acid; PN4, 6'-methyl paeonol; PN5, 3'-methyl paeonol; PN6, 2-methyl paeonol; Y, effective; N, not effective.

activity via blocking the function of NMDAR.^{24–26} The drug also prevents Aβ oligomers-induced depolarization²⁷ and neuronal death.^{28,29} Thus far, memantine is the only drug available that can attenuate NMDAR activity that is FDA approved.³⁰ This lack suggests that drug screening targeting this aspect of Aβ-induced pathophysiology is urgently needed, and thus strategies to search for such drugs that can ameliorate directly and/or indirectly the effect of Aβ oligomers are worth exploring.

Phytochemicals represent an outstanding resource for drug exploration. An extract from the root of the medicinal plant *Paeonia lactiflora* tastes sour; furthermore, it has been shown to modulate homeostasis.³¹ The extract, which contains the compound paeonol, has been investigated in relation to the treatment of AD, and the results mainly show amelioration of cognitive deficits in AD disease models.^{32–35} Thus far, there are few reports which develop the various paeonol derivatives to investigate their roles in the treatment of AD.^{36–38} In this study, we use a drug screening platform to search for herbal compounds that are able to ameliorate Aβ oligomers-induced pathophysiology. We targeted compounds extracted from *P. lactiflora* and found a paeonol derivative therefrom that has potential merit.

RESULTS

Synthesis of Paeonol Derivatives. Utilizing an established drug screening platform, we found that paeonol extracted from *P. lactiflora* is effective. The structure of

paeonol is shown in Figure 1A. The half maximal inhibitory concentration (IC₅₀) of paeonol based on the survival of glutamatergic neurons can not be determined thus far because the drug seems to have no effect even at 1 mM (Figure 1B). The effect of paeonol on AD has not been studied to a great extent and what research there is has mainly focused on its antioxidant effects^{34,35,39} and effects at the organism level.³³ We investigated this compound further by creating derivatives in addition to exploring its mechanism of action in more detail. Originally, we purchased the paeonol derivatives 4-methoxy salicylaldehyde (PN1), 4-methoxy salicylic acid (PN2), and 4-hydroxy salicylic acid (PN3), and their structures are shown in Figure 1C. However, we found these to be inactive using our drug screening platform.

The discrepancies in structure and function among paeonol and derivatives might be attributed to the existence and positioning of methyl groups. In light of this, we then synthesized paeonol derivatives by attaching methyl groups to different positions on paeonol and obtained 6'-methyl paeonol (PN4), 3'-methyl paeonol (PN5), and 2-methyl paeonol (PN6) (Figure 1C). As shown in Figure 1D, these paeonol derivatives PN4,^{40,41} PN5,⁴² and PN6^{43,44} were synthesized from resorcinol derivatives 1a–1c via Fries rearrangement/methylation reactions according to previously reported methods. The determination of composition and purity within organic synthesis is demonstrated by ¹H nuclear magnetic resonance (NMR) spectra (Figure S1). For the neuronal survival test, we adopted doses ranging from 30 to 500 μM. As

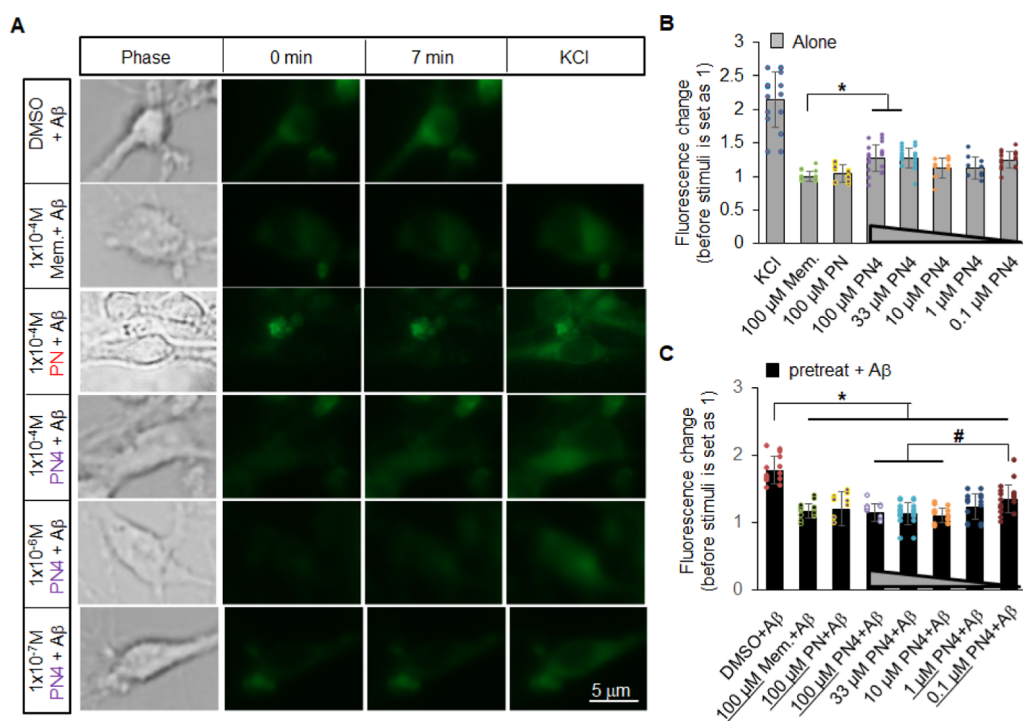


Figure 2. Both paeonol and 6'-methyl paeonol ameliorate Aβ-induced abnormal depolarization. (A) After cultivation for 6 days, primary cortical neurons were preincubated with different concentrations of 6'-methyl paeonol for a total of 2 h; during the last 0.5 h the medium was changed to KRH solution containing 0.8 μM DiBAC4(3). The change in fluorescence was recorded after the addition of 2.7 μM Aβ. In the experiments examining whether the drugs cause self-induced depolarization, these drugs were added directly to the neurons without preincubation. KCl 45 mM was added at 11 min at the end of the experiment to assess the responsiveness of the neurons. Memantine and paeonol 100 μM were included as a positive control and a comparison, respectively. (B,C) Images were analyzed and quantified using MetaMorph software. The change of fluorescent intensity across the different groups was plotted. [$F = 36.995$, $p < 0.001$, one-way ANOVA, $n = 15-20$]. * $p < 0.05$. PN, paeonol; PN4, 6'-methyl paeonol; Mem., memantine.

measured by the MTT assay, all of the paeonol derivatives except PN1 showed no influence on neuronal survival. That is, the neuronal viability demonstrated no significant decline even when the dosage reached 500 μM (Figure S2). As for the IC₅₀ of PN1, this was calculated to be 410 μM. Accordingly, we used concentrations much lower than 500 μM in all of the following experiments with all of the compounds. Among the synthesized compounds PN4~PN6, we identified two paeonol derivatives that were active on our platform, but we focused on PN4 in the follow-up studies (Figure 1E).

6'-Methyl Paeonol Ameliorates Aβ-Induced Abnormal Depolarization. As mentioned above, various other paeonol derivatives, having been either purchased or synthesized, were screened to determine their effects using our drug screening platform. Eventually, 6'-methyl paeonol was focused on. As shown in Figure 2A, preincubation with 6'-methyl paeonol over the range of 0.1–100 μM is able to block Aβ-induced depolarization. The effects of 100 μM paeonol and memantine were also examined, both as comparisons and controls. The fluorescence intensities using 6'-methyl paeonol alone at 0.1–100 μM were measured and quantitated. 6'-Methyl paeonol alone does not affect the membrane potential of cortical neurons except at higher doses. Treatment with 100 μM 6'-methyl paeonol alone slightly induced depolarization compared to the same dose of memantine, while low doses, such as 10 μM or less, did not have a detectable effect (Figure 2B). Aβ oligomers significantly increase the fluorescence intensity by about 1.5-fold to 2-fold. This increase could be prevented by preincubation with either memantine or 6'-

methyl paeonol at doses of 0.1–100 μM. These treatments showed a dose–response effect for 6'-methyl paeonol with respect to Aβ-induced depolarization (Figure 2C).

6'-Methyl Paeonol Ameliorates N-Methyl-D-aspartate and α-Amino-3-hydroxy-5-methyl-4-isoxazolepropionic Acid-Induced Neuronal Depolarization. We have previously shown that the abnormal depolarization caused by Aβ oligomers is mediated by receptors including NMDAR and AMPAR.²⁷ In order to dissect the mechanism of 6'-methyl paeonol mediating in the Aβ-induced pathophysiological state, we used agonists targeting those receptors to further study the pathway behind the activity. The images in Figure 3A show that preincubation with paeonol or 6'-methyl paeonol blocks NMDA-induced depolarization. The statistical results demonstrate that paeonol or 6'-methyl paeonol in the range of 0.1–100 μM has a significantly inhibitory effect (Figure 3B). An antagonist of NMDAR, MK801, was utilized to be a control, as it inhibits the NMDA-induced depolarization.

When the neurons were pretreated with 0.1–100 μM concentrations of 6'-methyl paeonol for 2 h followed by AMPA treatment, this derivative at all dosages inhibited the AMPA-induced depolarization, while paeonol pretreatment was only effective above 1 μM (Figure 4A). Quantification shows that 6'-methyl paeonol in the range of 0.1–100 μM is able to significantly reverse the AMPA-induced depolarization, but the 0.1 μM paeonol does not do this (Figure 4B). CNQX, an antagonist of AMPAR, was used as a positive control in the same experiment.

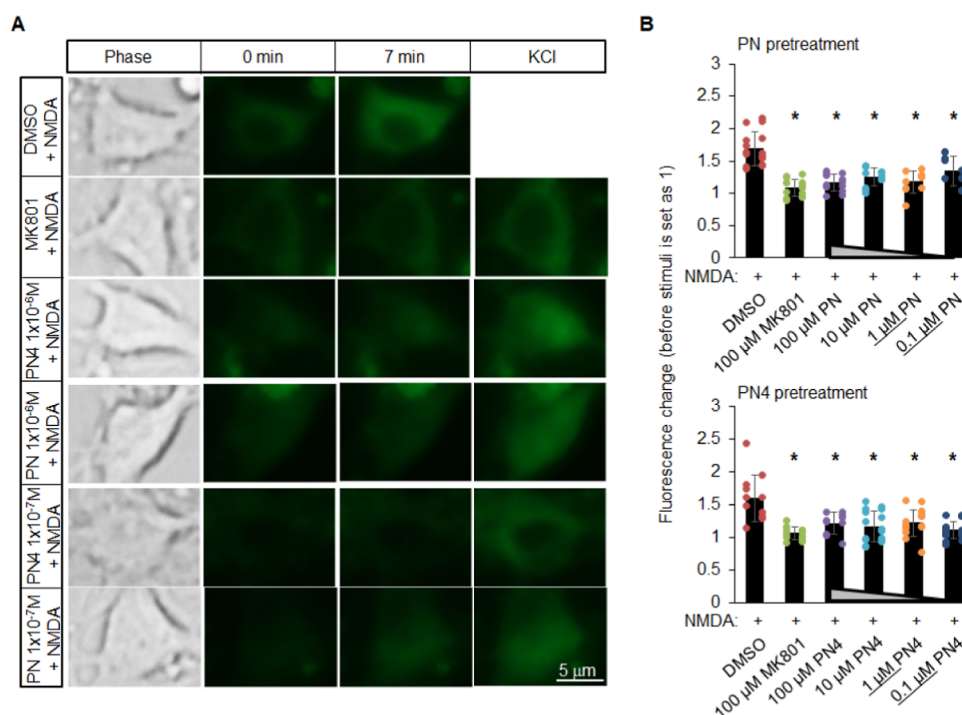


Figure 3. Both 6'-methyl paeonol and paeonol alleviate NMDA-induced depolarization at a similar concentration. (A) After cultivation for 6 days, primary cortical neurons were preincubated with different concentrations of 6'-methyl paeonol or paeonol for 2 h; during the last 0.5 h the medium was changed to KRH solution containing 0.8 μ M DiBAC4(3). Addition of 40 μ M NMDA initiated the neuronal depolarization. KCl 45 mM was added at 11 min at the end of the experiment to assess the responsiveness of the neurons. MK801, an inhibitor of NMDAR, was included as a positive control. (B) Images were analyzed using MetaMorph software. The changes of intensity within different concentrations of paeonol and 6'-methyl paeonol were quantified. [$F = 7.995$, $p = 0.002$, one-way ANOVA, and $n = 10-20$]. * $p < 0.05$ vs DMSO + NMDA.

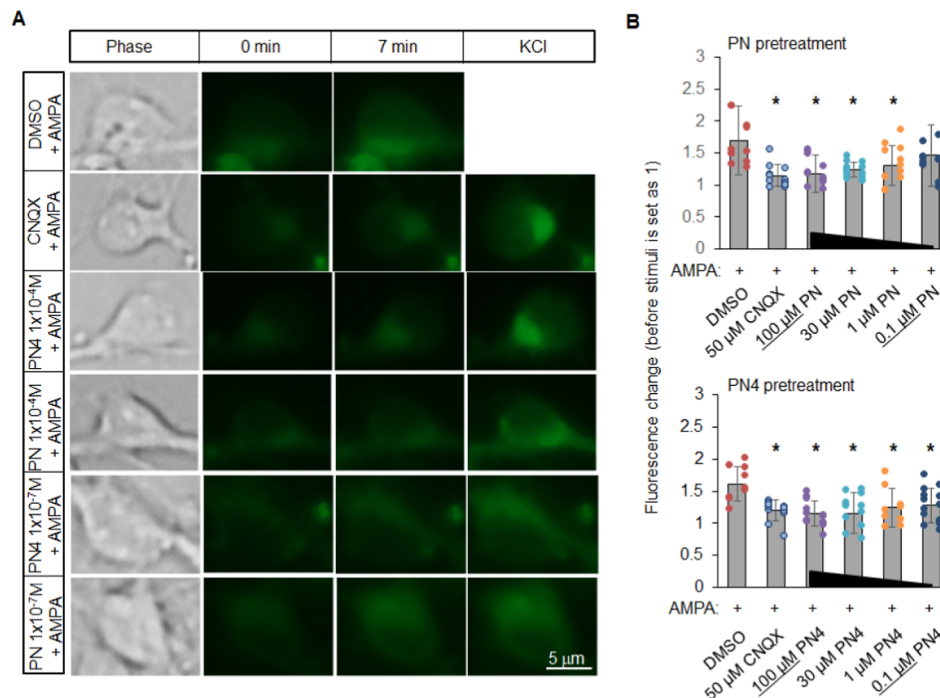


Figure 4. 6'-Methyl paeonol alleviates AMPA-induced depolarization at a lower concentration than paeonol does. (A) After being cultivated for 6 days, primary cortical neurons were preincubated with different concentrations of paeonol or 6'-methyl paeonol for a total of 2 h; during the last 0.5 h the medium was changed to KRH solution containing 0.8 μ M DiBAC4(3). Addition of AMPA 20 μ M initiated neuronal depolarization. KCl 45 mM was added at 11 min at the end of the experiment to assess the responsiveness of the neurons. CNQX, an inhibitor of AMPAR, was included as a positive control. (B) Images were analyzed using MetaMorph software. The changes of intensity at different concentration of paeonol and 6'-methyl paeonol were quantified. [$F = 9.963$, $p < 0.001$, one-way ANOVA, $n = 15-20$]. * $p < 0.05$ vs DMSO + AMPA.

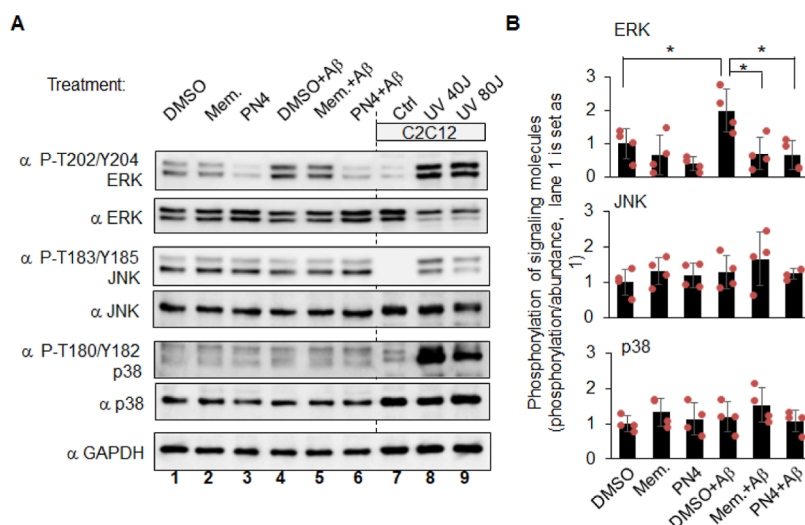


Figure 5. 6'-Methyl paeonol ameliorates the Aβ-induced increase in ERK phosphorylation. (A) After being cultivated for 6 days in vitro, the primary cortical neurons were preincubated with either vehicle or 100 μM 6'-methyl paeonol for 2 h, and then the cells were grown for 9 h with or without 2 μM Aβ. The changes in the signaling molecules were then analyzed by Western blotting. In the same experiment, pretreatment with 100 μM memantine was included as a control. The positions and sizes of the various signaling molecules were confirmed using UV irradiated C2C12 myoblasts. The presented blots are representative of an experiment that was repeated four times. (B) NIH ImageJ software was used to quantify the phosphorylation of various signaling molecules in the blots. The y-axis represents the ratio of the control values (lane 1), and this was set to 1. **p* < 0.05, two-tailed unpaired *t* test, *n* = 4. Mem., memantine; PN4, 6'-methyl paeonol.

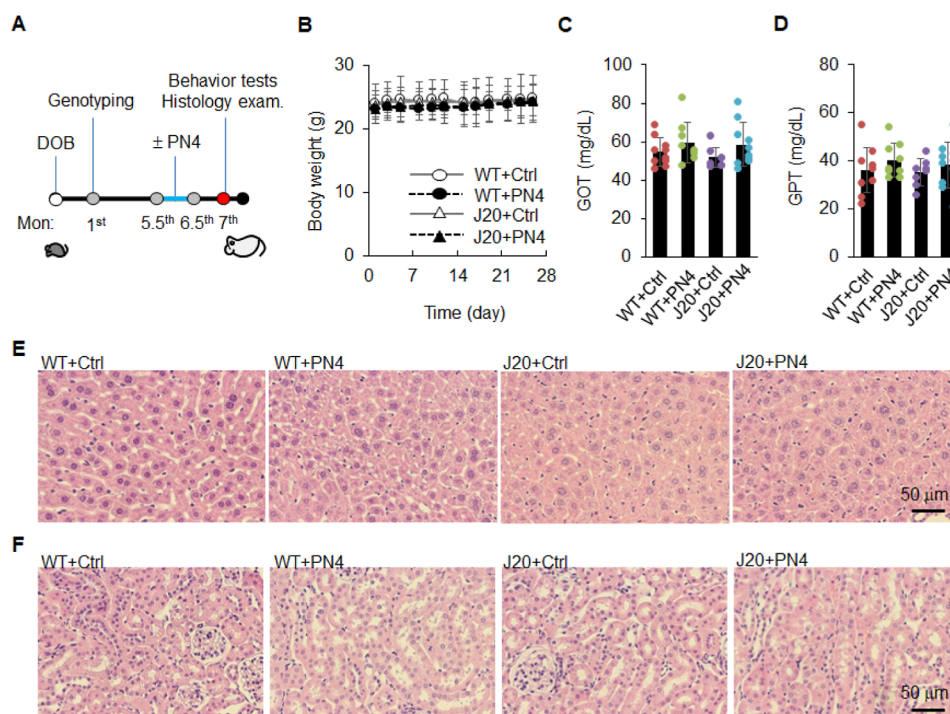


Figure 6. Effect of 6'-methyl paeonol on various mouse physiological parameters. (A) Flowchart of experimental procedures. (B) Body weight during the injection period of 6'-methyl paeonol was recorded and assessed. When the mice were around 7 months of age, the concentrations of (C) GOT and (D) GPT in the blood were measured. After sacrifice, hematoxylin and eosin (H&E) staining was done for (E) liver and (F) kidney tissue samples. Representative images are shown, *n* = 10 WT Ctrl, 9 WT PN4, 8 J20 Ctrl, 10 J20 PN4. WT, wild type; Ctrl, control; PN4, 6'-methyl paeonol.

Aβ-Induced Increase in ERK Phosphorylation is Prevented by 6'-Methyl Paeonol. We then focus on investigating the molecular mechanism of 6'-methyl paeonol. The signaling pathway activated by Aβ was examined. Both ERK and JNK are involved in Aβ-induced neuronal death.^{16,17} Moreover, Aβ-induced activation of p38 ultimately causes neuronal death via the activation of microglia during AD.¹⁸

Before the experiments, the optimal Aβ conditions were verified. As shown in Figure S3, we decided to use 2 μM Aβ to treat primary neurons for 9 h. The increase in ERK phosphorylation caused by Aβ was reversed by 6'-methyl paeonol, and this is similar to the effect of memantine (Figure 5A). The C2C12 cell samples (a mouse skeletal muscle cell line) were included in the analysis in order to confirm that the

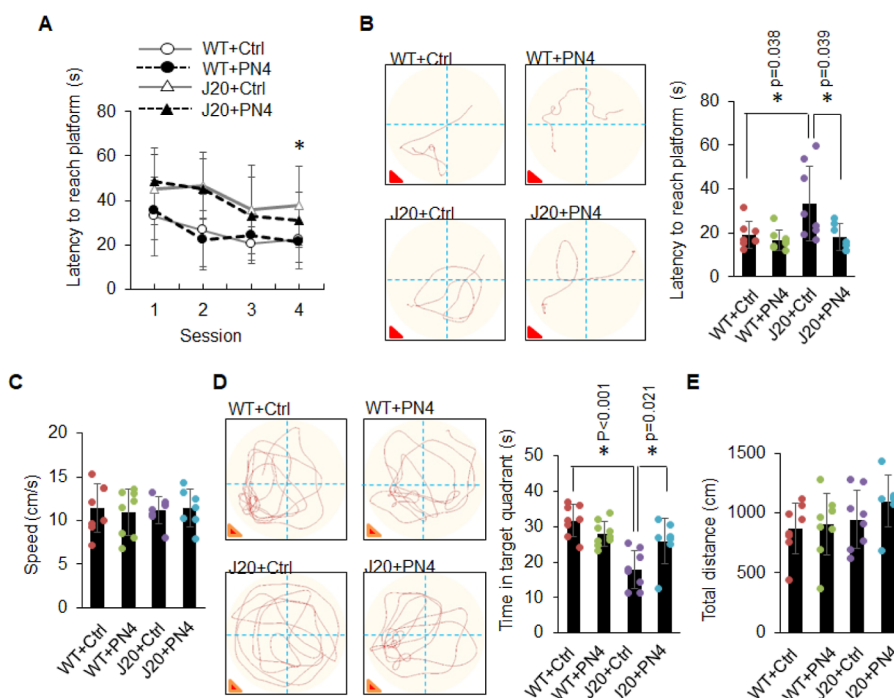


Figure 7. 6'-Methyl paeonol was found to ameliorate the defect in learning and memory behavior of the AD model J20 mice. (A) Mice were trained one session per day for 4 days in the Morris water maze test. The time each mouse needed to reach the hidden platform under water was measured during a training session. [Session 4, $F(1,33) = 6.682$, $p = 0.014$, two-way ANOVA, $n = 10$ WT Ctrl, 10 WT PN4, 9 J20 Ctrl, and 8 J20 PN4]. (B) Testing was performed a day after the final day of hidden-platform training. The time to reach the hidden platform was assessed. [$F(1,27) = 6.249$, $p = 0.019$, two-way ANOVA, $n = 8$ WT Ctrl, 8 WT PN4, 8 J20 Ctrl, and 7 J20 PN4]. (C) Speed to complete the testing was calculated and plotted. (D) 48 h after the testing, the hidden platform was removed. The mice were tested by measuring the time that they searched for the hidden platform within 1 min. [$F(1,27) = 12.608$, $p = 0.001$, two-way ANOVA, $n = 8$ WT Ctrl, 8 WT PN4, 8 J20 Ctrl, 7 J20 PN4]. (E) Total distances moved in the probe test are shown. All of the insets in the figure are representative traces. The red marks indicate the quadrant of platforms. WT, wild type; Ctrl, control; PN4, 6'-methyl paeonol.

correct signaling molecules were targeted, as shown in lane 7–9. Quantification of these blots was carried out via four independent experiments, and the results are presented in Figure 5B. The findings suggest that 6'-methyl paeonol significantly inhibits the A β -induced increase in ERK phosphorylation but has little or no effect on JNK/p38 phosphorylation.

Mice Treated with 6'-Methyl Paeonol Show No Effect on Various Physiological Parameters. We examined a number of parameters known to be associated with the basal level of toxicity that might affect the various experimental animals and tissues used in the present study. The presence of toxicity might influence the interpretation of the efficacy of 6'-methyl paeonol in our animal model. The experimental flowchart for these experiments is depicted in Figure 6A. At around 3 weeks of age, tail DNA from the wild-type and J20 mice was genotyped by polymerase chain reaction. These mice were injected with the vehicle or 6'-methyl paeonol 20 mg/kg body weight. This dose was based on a previous study of paeonol.³³ The animals received these treatments 5 days a week for 4 weeks when they were 5–6 months of age. These mice were used in the animal behavior experiments and also underwent hematological/histopathological examination at about 6–7 months old. The results for body weight (Figure 6B), glutamate oxaloacetate transaminase (GOT) activity (Figure 6C), and glutamic-pyruvic transaminase (GPT) activity (Figure 6D) showed that there were no significant differences between the treated and untreated groups. No identifiable abnormalities were observed when the histological

morphologies of the liver (Figure 6E) and kidney (Figure 6F) samples were examined.

6'-Methyl Paeonol Improves Impaired Learning and Memory Behavior of the J20 Mice. Initially, we examined the effect of 6'-methyl paeonol treatment on AD mice model J20 via the Barnes maze test, which is commonly used to assess the abilities of spatial learning and memory. There was no difference in the probe results among the four groups (Figure S4). Remarkably, the J20 model mice do not show the typical memory defect found in these mice when wild-type vs J20 mice are compared. Therefore, we also employ the Morris water maze test to examine the effect of 6'-methyl paeonol on spatial learning and memory based on the fact that using J20 mice is a well-established AD mice model that is suitable for the Morris water maze test.⁴⁵ During session 1, as shown in Figure 7A, all groups of mice at 7 months old are naive to the environment, and the latency for reaching the platform shows no significant differences between groups, although the wild-type groups do seem to be slightly faster overall than the J20 groups. In general, there seems to be a trend indicating that the latency for reaching the hidden platform is reduced across all groups of mice during training. Nevertheless, there are no statistical differences detected for this parameter until session 4 of training when the wild-type vehicle/PN4 injected mice and the J20 with vehicle injected mice were compared. Testing was performed at 24 h after the training period. The J20 mice with vehicle injection spent much more time than wild-type mice reaching the hidden platform, whereas the J20 mice with 6'-methyl paeonol injection reversed this behavior (Figure 7B).

The speeds in all groups are comparable (Figure 7C). 48 h after the testing, the probe was examined in which the hidden platform is removed. As seen in Figure 7D, the time spent in the target quadrant for the wild-type control mice group was significantly greater than the J20 control mice group. The 6'-methyl paeonol injected J20 mice also spent more time finding the platform in the target quadrant compared to the J20 control group (Figure 7D). However, the moving distance of the tested mice groups shows no significant differences in terms of their means, although there is a trend whereby the J20 + PN4 groups traveled a longer distance than the WT + Ctrl groups (Figure 7E). Since each group was measured over the same 1 min, the outcomes also indicate the speed of swimming. Hence, swimming ability does not seem to be affected by 6'-methyl paeonol nor by mice genotype. The results demonstrated that 6'-methyl paeonol alleviates the spatial learning and memory impairment of the J20 mice.

DISCUSSION

Paeonol has been shown to be able to inhibit oxidative stress-induced neuronal damage and microglia-mediated inflammation,³⁹ to mediate the RhoA/Rock2 pathway, attenuating the loss of neuronal dendritic spines³² and to increase the frequency of spontaneous excitatory postsynaptic currents and miniature excitatory postsynaptic currents.⁴⁶ Paeonol has also been found to inhibit the impairment of AD related behaviors in animal models.^{33–35} Our results on paeonol have not only verified the compound's AD-related activities, revealing more details regarding its molecular/cellular mechanism, but we have also showed that the compound has efficacy on our platform to prevent A β -induced depolarization. After modification of the structure of paeonol by adding methyl/hydroxyl groups, 6'-methyl paeonol is more effective than paeonol on the AMPA-induced abnormalities present in primary neurons. These findings suggest that 6'-methyl paeonol has potential as a useful candidate compound for treating AD.

ERKs have been shown to be crucial to learning and memory.⁴⁷ In particular, the ERK activation/pathway is involved in structural changes within dendritic spines and synapses.⁴⁸ Nevertheless, it has been suggested in this context that ERK1/2 plays a critical role in promoting neuronal death, which is an important aspect of many neurodegenerative diseases.⁴⁹ The persistent activation of ERKs is likely to mediate NMDA-related neurotoxicity.^{50,51} Therefore, both hyper- and hypo-activation of ERK might be involved in various disease pathways. Moreover, ERK activation has been shown to first increase and then later decrease in Alzheimer disease transgenic models. The amyloid precursor protein (APP) transgenic mice model involves a chronic activation of the ERK/MAPK pathway that eventually results in the downregulation of ERK2/MAPK and the reduction of CREB phosphorylation.⁵² In the same set of experiments, A β 42 was shown to be able to activate ERK in cultured rat hippocampal slices *in vitro*. In the brains of animals from the APP transgenic rat model that involves intraneuronal accumulation of A β peptides, a sustained activation of ERK has also been observed.⁵³ In the current study, we found that A β simply increases ERK phosphorylation over the time-course of the experiments. 6'-Methyl paeonol specifically reverses this A β -induced ERK phosphorylation. These findings not only show that 6'-methyl paeonol is able to ameliorate the early changes caused by A β , namely, increased ERK phosphorylation, but

also imply that this particular derivative might target signaling molecules within the ERK pathway. It is worth noting that A β actually causes activation/inhibition of a range of signaling molecules including caspases, JNK, p38, and others, in addition to ERK.¹⁴ However, we found that it is ERK that shows the most dramatic changes under the time and dose treatment with A β used here. Hence, 6'-methyl paeonol would seem to be able to ameliorate the A β -induced increase in ERK phosphorylation, at least in the model used here.

Based on the animal model results, the training behavior shows that the J20 mice injected with 6'-methyl paeonol display a trend toward spending less time finding the hidden platform during learning/training test day 4 compared to J20 mice with vehicle injection. This parameter is statistically significant, as shown in Figure 7B. This means the J20 mice injected with 6'-methyl paeonol were able to slowly learn and memorize how to find the platform, unlike the J20 mice with vehicle injection. Furthermore, the J20 mice injected with 6'-methyl paeonol lingered longer in the correct quadrant compared to the J20 mice injected with vehicle when the platform had been removed. This indicates that the impaired memory behavior in J20 mice seems to have been alleviated by treatment with 6'-methyl paeonol. Therefore, overall, these results show that 6'-methyl paeonol ameliorates spatial learning and memory impairment of J20 mice.

Recently, paeonol has been reported to be able to pass through the blood–brain barrier (BBB). Interestingly, it seems to be actively transported into the brain from the blood via a carrier-mediated transporter system.⁵⁴ Although we have not directly investigated whether or not 6'-methyl paeonol can pass through the BBB, the positive results of the behavior test imply that 6'-methyl paeonol does pass through the BBB and this derivative might utilize the same pathway or a similar pathway. Additionally, it is worth noting that 6'-methyl paeonol shows no effect when the Barnes maze behavioral test is used in this study. Specifically, there was no difference between WT and J20 mice at the beginning when this test was carried out. According to the literature, the Barnes maze seems not to be suitable for examining spatial learning and memory using the J20 mice model.⁴⁵ Thus, our results also confirm the above observation.

A comparison of structure and function in terms of the side groups on the benzene ring of paeonol and its paeonol derivatives is intriguing. The ability to alleviate A β -induced abnormal neuronal depolarization seems to be correlated with the compounds that contain methyl groups. This can be viewed in two ways. First of all, the effective derivatives on our screening platform show that the 1' position of the benzene ring of paeonol is largely in the form of ethanone and not in the form of benzoic acid benzaldehyde. This means that a 1' position methyl group might be effective in our assay. Second, the methyl group around the benzene might provide additional functionality in terms of its ability to inhibit A β -induced abnormalities. Whether the methyl group in this particular position is able to enhance the efficacy needs further study, including structural optimization. It has been shown that methyl groups might be able to provide/enhance noncovalent interactions as electron density donors and can form a hydrogen bond.⁵⁵ Furthermore, the two methyl groups could give rise to additional conformational changes. The effects of methylation of ligands in different combinations and positions on drug–kinase interaction have been studied previously. Interestingly, a single methyl group does not bring about

additive effects, while the presence of two methyl groups in one ligand can result in a novel ligand binding pattern within the complex of protein ligands.⁵⁶ Based on functional correlation and the structural similarities/differences between derivatives, we hypothesize that the ethanone at 1' and methyl groups at 3'4'6' of the benzene ring might be the areas to target when studying these functional groups, as well as the structural optimization of these compounds.

In summary, we found 6'-methyl paeonol, a paeonol derivative, is able to ameliorate A β -induced, NMDA-induced, and AMPA-induced depolarization in primary cortical neurons. The compound is also able to inhibit the increased phosphorylation of ERK induced by A β . At the organism level, 6'-methyl paeonol alleviates the defect of spatial learning and memory in J20 mice. On the basis of structural comparisons, the presence of a methyl group or groups on the benzene ring of paeonol derivatives might be useful when developing more effective candidate drugs. Our findings provide crucial information for the further structural optimization of paeonol-derived drugs for treating AD.

MATERIALS AND METHODS

Reagents and Antibodies. The reagents and antibodies used in this study included Neurobasal-A medium, L-glutamine, B27 supplement (Invitrogen/Gibco, Grand Island, NY, USA), 3-(4,5-dimethyl-2-thiazolyl)-2,5-diphenyl-2H-tetrazolium bromide (MTT) (Sigma, St. Louis, MO, USA), β -amyloid 1–42 (AnaSpec, Fremont, CA, USA), antibodies against ERKs/MAPK p42/44, ERKs/MAPK p38/44, Thr202/Tyr204, JNK Thr183/Tyr185, p38, p38 Thr180/Tyr182 (Cell Signaling Technology, Danvers, MA, USA), JNK, GAPDH (abcam, Cambridge, UK), and horseradish peroxidase (HRP) conjugated goat antimouse secondary antibody, and goat antirabbit secondary antibody (Jackson ImmunoResearch, West Grove, PA, USA).

We purchased commercially available paeonol (catalog no. sc205787, Santa Cruz Biotechnology, Santa Cruz, CA, USA) and paeonol derivatives 4-methoxy salicylaldehyde (PN1) (catalog no. AR-17108, Acros Organics, Thermo Fisher Scientific, Waltham, MA, USA), 4-methoxy salicylic acid (PN2) (catalog no. AF-H55345, Alfa Aesar, Thermo Fisher Scientific, Haverhill, MA, USA), and 4-hydroxyl salicylic acid (PN3) (catalog no. AR-11487, Acros Organics, Thermo Fisher Scientific, Waltham, MA, USA). All other chemicals were purchased from Sigma (St. Louis, MO, USA) unless indicated.

Experimental Animals and Design. All animal experiments were performed with the approval of the Institutional Animal Care and Use Committee of National Taiwan Normal University (NTNU permission no. 108038). C57BL/6 mice were purchased from the Academia Sinica Animal Center (Taipei, Taiwan). The APP transgenic mice (line J20) carried the human APP gene with the Swedish (K670N/M671L) and Indiana (V717F) familial mutations and were used as the AD mouse model. Mice were housed in a light/dark cycle of 12 h light and 12 h dark. Food and water for mice were provided *ad libitum*.

The experimental design for the behavior tests was as follows. After weaning at 3 weeks of age, the wild-type and J20 mice were genotyped using tail DNA and the probes recommended by Jackson Laboratory. The mice were grouped into WT vs J20 mice and given subcutaneous injections of either vehicle [mineral oil containing 20% dimethyl sulfoxide (DMSO)] or 6'-methyl paeonol dissolved in vehicle. There were 10 mice in each group. These mice were injected 5 days a week for 4 weeks when they reached an age of 5.5–6.5 months old. At this point the mice were subjected to the animal behavior tests, and this was followed by hematological/histopathological analysis at about 7 months old.

Mouse Primary Cortical/Glutamatergic Neurons. After mating and delivery, mice at postnatal days 1–3 (P1 to P3) were sacrificed and dissected to obtain dissociated cerebral cortex. The

protocol used was adopted from Lin et al.²⁷ In brief, the cortex tissue was digested with papain. The isolated cells were then centrifuged, and the supernatant was collected. Next these cells were carefully laid onto a prepared OptiPrep gradient in a centrifuge tube. After centrifugation, cells were collected from the lower layer. These neuronal cells were plated onto poly-D-lysine coated culture dishes. Around 90% of the cellular population is made up of glutamatergic neurons, and this was verified using vGLUT1/2 and Type III tubule antibodies.

3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium Bromide (MTT) Assay. Cell viability was assessed by the MTT assay as described previously.^{27,57} In brief, the tetrazolium dye MTT was incubated with the neurons. The medium was replaced with DMSO in order to dissolve the formazan crystals. Absorbance was measured at 570 nm by using a microplate reader. Absorbance is presented as a percentage of the vehicle control.

Measurement of Neuronal Depolarization. The measurement of neuronal depolarization, along with a drug screening platform, was developed and established previously in our laboratory.^{27,57} In brief, the medium of primary neurons was changed to KRH buffer [127 mM NaCl, 5 mM KCl, 2 mM CaCl₂, 1.2 mM MgCl₂, 6 mM glucose, and 25 mM 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES)]. Reagents such as DiBAC4(3), herbal extract, or pure compound, together with 4',6-diamidino-2-phenylindole (DAPI) or Hoechst 33342 were added to the KRH buffer. After buffer change/dye incubation, neuronal images were recorded using inverted fluorescent microscopy (Carl Zeiss, Jena, Germany). Specifically, A β 42 oligomers preparation and formation were assessed as described previously and verified using 6E10/A11 antibodies.²⁷ On the basis of nucleus staining as the reference for cell position, the changes in green fluorescence were quantified using MetaMorph software. At least 15–30 neurons were quantified from each sample.

Western Blot Analysis. This was carried out as described previously.^{27,57} In brief, the neuronal cells were lysed in lysis buffer consisting of 20 mM Tris-Base (pH 7.9), 20 mM NaCl, 1 mM EDTA, 20 mM β -glycerophosphate, 1 mM DTT, 1 mM PMSF, 25 mM calyculin A, 1 tablet/50 mL of protease inhibitor (Roche, Mannheim, Germany), and 0.5% Triton X-100. After sodium dodecyl sulfate–polyacrylamide gel electrophoresis analysis and PVDF membrane transfer, the blots were visualized using an appropriate HRP-conjugated secondary antibody, followed by visualization using a chemiluminescence kit (Millipore Corporation, Billerica, MA, USA).

Treatment of Mice and Mouse Behavior Testing. Since 6'-methyl paeonol is a derivative of paeonol, a 20 mg/kg body weight dose was injected into the two groups of mice based on a previous report in which paeonol was used. This dose was also used for the animal behavior testing.³³ It is worth noting that 6'-methyl paeonol, which has a molecular weight of 180.203 g/mol, was initially dissolved in DMSO to give a 20 mg/mL stock solution. Being aware of the toxicity of pure DMSO, we used a 4-fold volume of mineral oil to dilute the stock solution before subcutaneous injection into the mice. The final concentration of DMSO in each subcutaneous injection used for *in vivo* testing was thus 20%, and the injection volume of mineral oil plus DMSO was approximately 110 μ L for a 22 g body weight mouse. After the behavior tests, the mice underwent urethane or Zoletil anesthesia and then were transcardially perfused with saline followed by 4% paraformaldehyde. After this was completed, the livers and kidneys of the mice were removed, postfixed, and examined histologically. The measurements carried out to assess the hematology, gross pathology, and histopathology of the mouse samples were performed by the Mouse Core Facility (Academia Sinica Mice Clinical, Taipei, Taiwan). The replicate number for each experimental group is stated alongside the number of replicates that form each group. Some GOT/GPT data are absent because the level of hemolysis within samples is too high to be reliable.

Barnes Maze. The test used the parameters previously reported in the literature.⁵⁸ In brief, this maze contains 20 holes around the perimeter of a 120 cm diameter white polyethylene platform elevated 50 cm from the floor to the roof. Three 100 W flood lamps and a CCD were placed above the maze; the former were used to act as an

aversive stimulus, while the latter was used to track the location of each mouse. At the beginning of the habituation/pretraining phase, the mice were placed in the center of the apparatus covered by a box for 1 min; this was followed by removal of the box to allow exploration for 1 min to find an escape hole/box. Mice were guided toward the escape hole/box by a piece of food if they did not enter the escape hole/box within the above time. This pretraining phase consisted of only one trial per mouse on a single day. The next phase was the acquisition training phase, in which the mice were given 3 min to locate the escape hole/box. This training phase consisted of three trials per mouse each day for 4 days. The table and escape hole/box was cleaned between trials to prevent odor cues from developing on the platform. The day after the acquisition training phase, the mice were given a 3 min probe trial. During this trial, the escape hole/box was removed. Two days after the first probe trial, the mice were given a 3 min probe trial again. The latency to enter the escape hole/box, their average speed, and the distance traveled were collected and analyzed for each trial using EthoVision software (XT 17, Noldus, Wageningen, Netherlands).

Morris Water Maze. The protocol used was adopted from Yeh et al.⁵⁹ Briefly, a circular platform 9 cm in diameter was submerged in a water pool that was 120 cm in diameter. Hidden-platform training, testing, and probe trials were then conducted. Their time to reach or search for the platform and their swim speed or distance were recorded and analyzed by using a video tracking system and EthoVision software (XT 17, Noldus, Wageningen, Netherlands). It is worth noting that some sample sizes were reduced by 1 or 2 individuals when the Morris water maze test was performed due to the mice not swimming at all but rather just randomly drifting around in the water during measurement. The number of groups is stated alongside the number of replicates in each group.

Statistical Analysis. The Statistical Package for the Social Sciences (SPSS) software package was used to perform various statistical analyses. Differences among multiple means were assessed by one-way or two-way analysis of variance (ANOVA), as appropriate, and this was followed by the Dunnett's *post hoc* test. Some treatment effects were also evaluated using the two-tailed Student's *t*-test and Excel software. Data are presented as the means $\pm$ SD. A *p* value of <0.05 was considered to be statistically significant.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acschemneuro.3c00633>.

¹H NMR spectra of PN4–PN6 synthesis; effect of PN1–PN6 on neuronal survival; optimal treatment of A β conditions; effect of PN4 treatment on mice via the Barnes maze test; and all full blots shown in Figure 5 (PDF)

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Author Contributions

[§]P.J.C. and C.A.Y. contributed equally to this work. P.J.C., C.A.Y., P.C.C., H.J.T., Y.R.C., and J.H.C. designed and performed the experiments. P.L.C. and G.C.L. provided essential materials. W.L. and Y.L. designed the experiments, analyzed the data, and wrote the paper. All authors have approved the final manuscript.

Notes

The authors declare no competing financial interest.

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